Final Project 5450 - Exploration of Modeling in Game Theory

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1 Introduction

The intent of this paper is to an introductory exploration of game theory. Extensive study has been done on the modeling of cooperative and non-cooperative human-interaction-dynamics as influenced by behaviorally reinforcing constructs. This research, at a high level, is referred to as game theory. Game Theory has been shown to prove useful in myriad domains. A non-comprehensive list including subject matter from Evolutionary Biology, Economics, Business and various Social Studies (military, diplomacy etc...,). All these efforts predominantly in the vain of predicting what "someone" else may do. In the case of evolutionary studies, or disease spread, modeling expands into examining how living things behave under iterated steps of change as modeled under the influence of "game" rules and strategies. Mathematicians & Economists began applying the tools at their disposal to model what might influence a game's outcome.
An early examination by John Von Neumann and Oskar Morgenstern *Theory of Games and Economic Behavior*, first published in 1944 has extensive documentation of zero-sum cooperative games from 2 to n ≥ 5 players. In 1949 John Nash published his PhD dissertation entitled "Non-Cooperative Games" in which he discusses non-cooperative games and the ideas of equilibrium. The Nash Equilibrium is when no player can improve their outcome by a move of their own alone. A well known type of game that is commonly used as an introduction to the concepts of game theory is called the Prison Dilemma. There are countless variations on the study of these game modifications and the resulting outcomes. The set up of the game allows researchers to dial the parameters of engagement such that many different salient questions about strategy and outcomes can be answered. We will brush the surface of game theory using variations of the prisoner's dilemma games. We will start with the basic single-shot game, followed by an examination of pure strategies & mixed strategies. Finally we will wrap with a brief look at how replicator dynamics influence the outcomes.

2 Game Theory

2.1 The Prisoner's Dilemma

We will begin our introduction to game theory with the basic structure of the classic Prisoners Dilemma Game. In 1952 a paper, *Some Experimental Games*, commissioned by the military out of the RAND corporation written by Merrill M. Flood, discusses experimentation in game theory beyond the theory understood at the time. Inspired by both Von Neumann & Morgenstern and Nash, he and his collaborators tested the outcomes over varying game setups over several years of experimentation.
One of those experiments ran in 1950, done in collaboration between Flood and Dresher. This section of Merrill's paper examined the behavior of a non-cooperative pair. They were testing whether a "two-person positive-sum non-cooperative game"[2] setup when run over several rounds would have a Nash Equilibrium when administered in a real-world environment. Their payoff matrices were constructed with a specific single Nash Equilibrium. Their analysis examined the mathematical outcomes. In an attempt to make the implications of the outcome more understandable to people in other fields of research Albert Tucker wrapped the math with an easy to understand story of the Prisoners dilemma. The Stanford Encyclopedia of Philosophy contains a very concise synopsis of the setup of The Prisoner's Dilemma directly qouted below:[4]

> Tanya and Cinque have been arrested for robbing the Hibernia Savings Bank and placed in separate isolation cells. Both care much more about their personal freedom than about the welfare of their accomplice. A clever prosecutor makes the following offer to each: "You may choose to confess or remain silent. If you confess and your accomplice remains silent I will drop all charges against you and use your testimony to ensure that your accomplice does serious time. Likewise, if your accomplice confesses while you remain silent, they will go free while you do the time. If you both confess I get two convictions, but I'll see to it that you both get early parole. If you both remain silent, I'll have to settle for token sentences on firearms possession charges. If you wish to confess, you must leave a note with the jailer before my return tomorrow morning."

Flood & Dresher 2x2 Matrix Payoff Matrix

AA=

	1	**2**
1	-1	$\frac{1}{2}$
2	0	1

JW=

	1	**2**
1	2	1
2	$\frac{1}{2}$	-1

The Nash Equilibrium for this payoff matrix is when AA plays 2 and JW plays 1. AA is the row player and makes decisions applied row-wise. JW is the column player and makes decisions applied column wise. If AA is deciding what to play he looks at what JW may do. If JW plays 1, than AA would be better to play 2 and get 0 rather than -1. If JW plays 2, AA would be better to play 2 and get 1 rather than $\frac{1}{2}$. For JW he is considering what AA may do to surmise his best choice. If AA plays 1, JW is better to play 2, he would receive 2 rather than 1. If AA plays 2, JW is better off playing 2 again as he receives $\frac{1}{2}$ versus -1. In this match up the Nash Equilibrium is (2,1). After 100 turns the resulting payoff, had the Nash Equilibrium held, would have a payout of 0 to AA and .5 to JW. This was not what was observed in experimentation. Over several rounds of play they found that this payoff to players was never achieved. This game was a generalized form of non-cooperative play. Next we will look at prisoners dilemma specifically.
The most basic standard form of Prisoners dilemma consists of one turn played between 2 players.

Standard Prisoners Dilemma Payoff Matrix

Player 1=

	C	**D**
C	3	0
D	5	1

Player 2=

	C	**D**
C	3	5
D	0	1

$$T > R > P > S$$

$$5 > 3 > 1 > 0$$

Player 1 is the "row" player. Player 2 is the "column" player. There are 2 strategies available in this set up, cooperation and defection. This can be represented by $\sigma = (\sigma_1, ...\sigma_n)$ where n is the total number of strategies being played. In our example there are only 2 strategies, σ_1 & σ_2. For simplicity of discussion in this 2 strategy situation, we will represent the 2 strategies by C and D, cooperation and defection respectively. The possible combination of plays in the single interaction between player 1 (the first letter) and player 2 (the second letter) are:

$$(C, C), (C, D), (D, D), (D, C)$$

Given that the unknown to any given player is their opponents strategy choice, we analyze what the opponent may choose to do. For player 1, she will consider what player 2 strategy options are and how that would affect her own payoff (vice versa for player 2).

Player 2 chooses strategy C, player 1 best choice: $T_r > R_r$
Player 2 chooses strategy D, player 1 best choice: $P_r > S_r$
Player 1 chooses strategy C, player 2 best choice: $T_c > R_c$
Player 1 chooses strategy D, player 2 best choice: $P_c > S_c$

Each players best choice in response to *any* potential opponent choice is for them to choose defection. This is the contradiction that results. Rational individuals will choose to defect even though choosing to cooperate would result in overall communal good. More specifically:

The mathematical representation for strategy(σ_i^*) to be a Nash Equilibrium:

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*), \quad \forall i, \forall \sigma_i$$

where σ_{-i} represents all other opponent strategies. Many references to the Nash Equilibrium will characterize this in words as when a player cannot improve their payoff by himself by choosing another strategy. In this

case the Nash Equilibrium is (D, D) where both players choose to defect. Defection is the best choice for any player regardless of what their opponent(s) choose. If the play follows its Nash Equilibrium then both players would always defect and get a payout of 1 each, for a total payout of 2.

This 2x2 single-shot play provided the foundation which then led to extensive analysis of more complicated game strategies with more players. We will look at some of the math utilized in more complicated games, within the construct of this simpler scenario.

2.2 Notation

Before we begin the mathematical introduction it is worth taking a moment to discuss notation. Game Theory spans many domains of study. As a result there several different norms of language and variable representation used when applying game theory. For the purposes of this discussion we will explicitly be using the below.[9]

$$I = \{1, 2, \ldots, n\} \text{ The Player Set}$$

Playing Pure Strategy:

$$\begin{aligned} S^i &= \{e_1, e_2, \ldots e_{m^i}\} \text{ Pure Strategy Set for Each Player } i \\ s &= (s^1, s^2, \ldots s^n) \text{ Pure Strategy Profile} \\ S &= S^1 \times S^2 \times \cdots \times S^n \text{ Pure-Strategy Profile Space} \\ u^i(s) &\text{ Payoff to Player } i \text{ when profile } s \text{ is played} \end{aligned}$$

Playing Mixed Strategy:

$$\begin{aligned} \sigma^i &= (\sigma_1^i, \sigma_2^i, \ldots, \sigma_{m^i}^i) \in \Delta(S^i) \text{ Mixed Strategy for Player } i \\ &= (\sigma_1^i \cdot e_1 + \sigma_2^i \cdot e_2 + \ldots + \sigma_{m^i}^i \cdot e_{m^i}) \\ \sigma &= (\sigma^1, \sigma^2, \ldots, \sigma^n) \text{ A Mixed-Strategy Profile} \\ \sigma &\in \Delta(S^1) \times \Delta(S^2) \times \cdots \times \Delta(S^n) \\ U^i(\sigma) &= \text{ Expected Payoff for Player } i \text{ when mixed-strategy profile } \sigma \text{ is played} \end{aligned}$$

2.3 Pure Strategy

A strategy which is defined strategy selected where a player as a set method for selecting a strategy. Examples include "Always Cooperate", "Always Defect", "Tit For Tat". Here we will assume a player does not switch strategies mid round.
To calculate the payoff for each player:

- Always Cooperate: This player intentionally will always pick Cooperate. Her strategy set as defined can only include:
$$S^1 = \{1\}$$
- Always Defect: This player intentionally will always pick Defect. Her strategy set as defined can only include:
$$S^2 = \{2\}$$
- Tit For Tat: This player starts with Cooperate and copies their opponent everytime thereafter *(there are many variations on this strategy)*. His strategy set can include choosing either of our 2 strategies:
$$S^3 = \{1, 2\}$$

We are examining a few 25-game match ups below.

2.3.1 Cooperator vs. Defector

In this case player 1 chooses to always cooperate and player 2 chooses to always defect. A highly unrealistic game in that in the face of continual loss, player 1 stays with her strategy of cooperate.

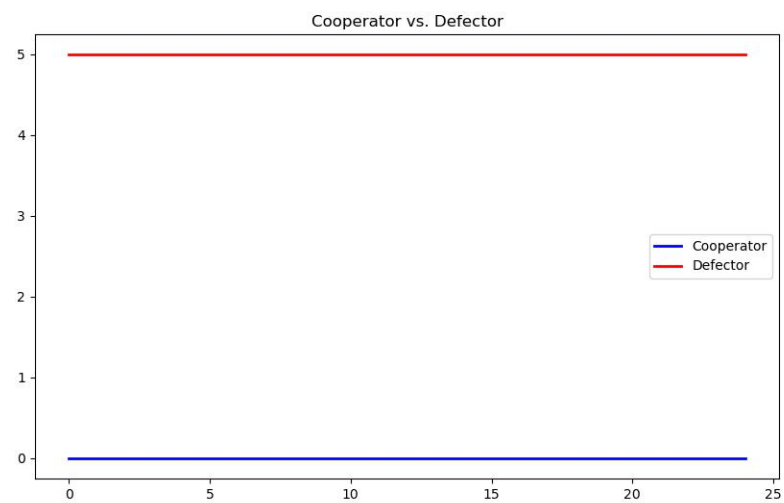
$$u^1_{s_1}(s_1, s_2) = e^T_{s_1} A e_{s_2}$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

$$u^2_{s_2}(s_1, s_2) = e^T_{s_2} A e_{s_1}$$

$$= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 5$$

Here we see the math agrees. Player 1 always cooperating will get 0 payout and Player 2 will get 5 each match.



2.3.2 Cooperator vs. Tit For Tat

With Cooperator playing against Tit For Tat we find that Tit For Tat "plays nicely" resulting in equal payouts for the players. Our Tit For Tat player starts out cooperating and never changes. Effectively emulating Always Cooperator.

$$u^1_{s_1}(s_1, s_3) = e^T_{s_1} A e_{s_3}$$

Here we must ask ourselves, what does s_3 represent? Can this be calculated with our current methods for calculating expected payout? Strategy s_3 allows player 3 to select either pure strategy of cooperate (1) or defect (2) given a certain set of rules that cannot be modeled here over 25 matches. We can calculate the first match manually by following the rules of the strategies for each player. Because player 3 starts as a cooperator and player 1 is always a cooperator for our first match up we arrive at:

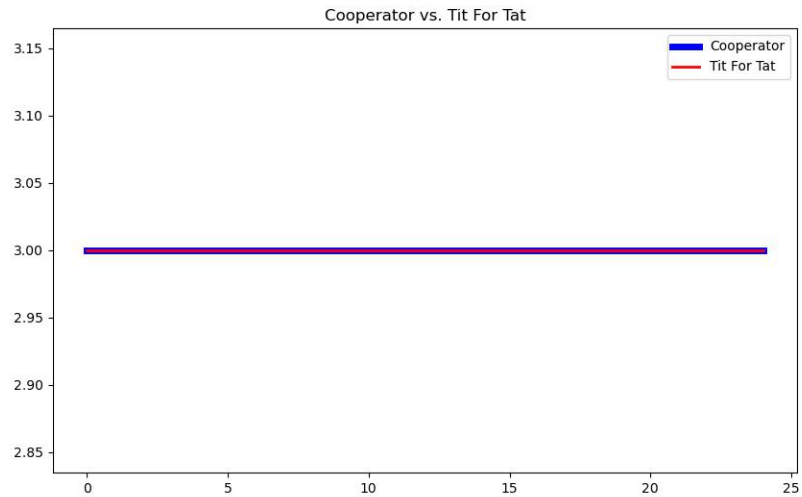
$$u^1_{s_1}(s_1, s_1) = e^T_{s_1} A e_{s_1}$$

$$u^1_{s_1} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 3$$

$$\text{and } u^3_{s_1}(s_1, s_1) = e^T_{s_1} A e_{s_1}$$

$$u^3_{s_1} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 3$$

If we were to manually follow the tit for tat logic through the entirety of the game we would find that both players receive 3 each match.



This results in the highest group payout when summing the accumulation of payouts to both players.

2.3.3 Defector vs. Tit For Tat

Similar to our prior discussion on how to model Tit For Tat we are going to need to manually look at the behavior of the players for the first match. Here it makes to the also look at the second match when our player 3 emulates our player 2's strategy.

$$u^2_{s_2}(s_2, s_1) = e^T_{s_2} A e_{s_1}$$

$$u^2_{s_2} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 5$$

$$u^3_{s_1}(s_2, s_1) = e^T_{s_1} A e_{s_2}$$

$$u^3_{s_1} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

Our Tit for Tat player 3 will not let that happen again and copies our player 2 defector. This pattern continues for the remainder of the matches.

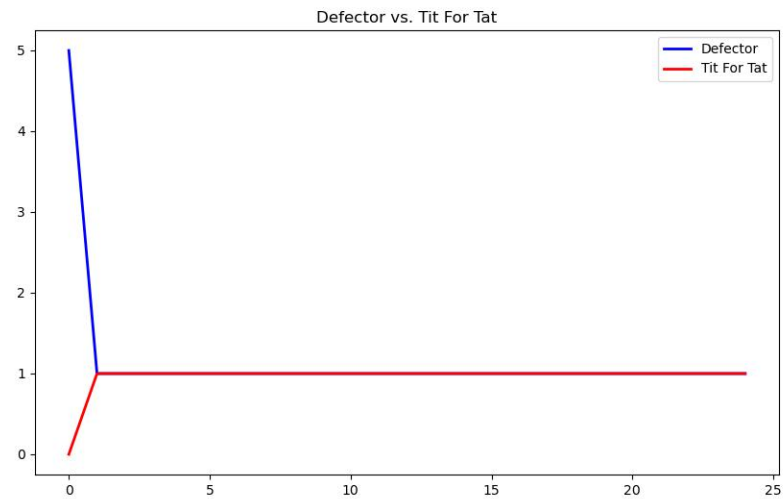
$$u^2_{s_2}(s_2, s_2) = e^T_{s_2} A e_{s_2}$$

$$u^2_{s_2} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

$$u^3_{s_2}(s_2, s_2) = e^T_{s_2} A e_{s_2}$$

$$u^2_{s_2} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

For the remainder of the matches each player will receive 1.



We can see with these fairly basic strategies the highest communal payout results if both players cooperate. The best individual payout can only be guaranteed if a player defects. Where best is defined as the strategy that cannot best itself, by itself. Because we assume we have rational players who strategize based on knowing the payoff to both players, we will see that both players receive less than if they had cooperated.

2.4 Mixed Strategy

At this point we can turn our attention to slightly more interesting dynamics in the iterated prisoners dilemma. The rules of "opportunities to switch" and "how & why to switch" are left for more significant works. Here we will simply say players have the opportunity to swap from one pure strategy to another at any given round. This is represented on the whole of several rounds of matches as a probability distribution over the strategy set of a player. Many researchers are interested in motivation and causality of strategy decision-making. We are not so much interested in the human element of why a player may choose a particular strategy, or why evolutionary processes include changes in strategies, both having no apparent best strategy. We are here to look at outcome results that can be mathematically modeled.

Returning to our 2-strategy 2-player set up of prisoners dilemma with the following pay off matrix.

A=

	C	**D**
C	3	0
D	5	1

In this case where the players face-off over several matches their strategy switching is modeled by a probability distribution. We can think of this as the probability of a player selecting a particular strategy at a given match, or the expected proportion of payoffs that will result for a player based on their mixed strategy set. The math is similar to the pure strategy math with a simple weighting for payouts by probability pairs of a

particular mix between both players.
Expected Payoff payoff for each player is given by the following formula:

$$U^1(\sigma^1, \sigma^2) = (\sigma^1)^T A\sigma^2, \quad \text{and} \quad U^1(\sigma^2, \sigma^1) = (\sigma^2)^T A\sigma^1$$

$$\begin{aligned}\sigma^i &= (\sigma^i_1, \sigma^i_2, \ldots, \sigma^i_{m^i}) \in \Delta(S^i) \text{ Mixed Strategy for Player } i \\ &= (\sigma^i_1 \cdot e_1 + \sigma^i_2 \cdot e_2 + \ldots + \sigma^i_{m^i} \cdot e_{m^i})\end{aligned}$$

$$\sigma^1 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} \qquad \sigma^2 = \begin{bmatrix} .1 \\ .9 \end{bmatrix}$$

The expected payoff for a player is the product of her strategy matrix multiplied with the payoff matrix weighted by her opponents strategy matrix. Analytically this set up can be modeled by:

$$U^1(\sigma^1, \sigma^2) = \begin{bmatrix} .6 & .4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} .1 \\ .9 \end{bmatrix} = \begin{bmatrix} .6 & .4 \end{bmatrix} \begin{bmatrix} 0.3 \\ 1.4 \end{bmatrix} = (0.6) * (0.3) + (0.4) * (1.4) = 0.74$$

$$U^2(\sigma^2, \sigma^1) = \begin{bmatrix} .1 & .9 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} .6 \\ .4 \end{bmatrix} = \begin{bmatrix} .1 & .9 \end{bmatrix} \begin{bmatrix} 1.8 \\ 3.4 \end{bmatrix} = (0.1) * (1.8) + (0.9) * (3.4) = 3.24$$

Based on this calculation *overtime* we would expect player 1 payoff to be ≈ 3.24 and player 2 ≈ 0.74. In words, the summation of the payoff matrix multiplied by the probability pairs of the players strategies.
Many times in the examination of particular processes or behaviors we are not just interested in the expected payoffs at a stable equilibrium, assuming there is one. We frequently want to examine within game dynamics. We ask what player behavior occurs during the round over several matches? Again *why* is not of concern in our discussion.
Lets say we took the perspective that at each match the payoff matrix σ^i may be adapted by each player i in accordance with some defined rules appropriate for a given game context. Perhaps the player might imitate players who are successful, or will cooperation emerge where communal payout begin to dominate the outcomes? At this point we turn our attention to one of the most important formulas in Game Theory. It is referred to as the Replicator Dynamic.

2.5 Taylor-Jonker Replicator Dynamic

Here is where the proverbial rubber meets the road for what can be accomplish in dynamic modeling of game outcomes. In studying single species evolution dynamics, Taylor and Jonker [1, 7] developed what came to be referred to as the replicator equation.

$$\dot{\sigma_h}^i = \sigma^i_h \left[u^i_h(\sigma^j) - \bar{u}^i(\sigma^j) \right]$$

In words: the rate of change of a player i's strategy mix of a given strategy h can be given by the difference between the payoff of strategy h by player i and the average payoff of player i against player j's entire strategy set. This average being calculated by the weighted sum of each payoff pair between the players. The difference then being multiplied by the probability that player i would even play strategy h.
An example in our 2 player case:
For any given payoff matrix a characteristic continuous ODE can be derived as follows for a mixed strategy

game:

$$\begin{aligned}
\dot{\sigma_h}^1 &= \sigma_h^1 \left[u_h^1(\sigma^2) - \bar{u}^1(\sigma^2)\right] \\
&= \sigma_h^1 u_h^1(\sigma^2) - (\sigma_h^1 u_h^1(\sigma^2) + \sigma_k^1 u_k^1(\sigma^2)) \\
&= \sigma_h^1 u_h^1(\sigma^2) - \sigma_h^1 \sigma_h^1 u_h^1(\sigma^2) - \sigma_h^1 \sigma_k^1 u_k^1(\sigma^2) \\
&= u_h^1(\sigma^2)[\sigma_h^1 - \sigma_h^1 \sigma_h^1] - \sigma_h^1 \sigma_k^1 u_k^1(\sigma^2) \\
&= u_h^1(\sigma^2)\sigma_h^1[1 - \sigma_h^1] - \sigma_h^1 \sigma_k^1 u_k^1(\sigma^2) \\
&= u_h^1(\sigma^2)\sigma_h^1 \sigma_k^1 - \sigma_h^1 \sigma_k^1 u_k^1(\sigma^2) \\
&= \sigma_h^1 \sigma_k^1 (u_h^1(\sigma^2) - u_k^1(\sigma^2))
\end{aligned}$$

For our specific payoff matrix:

$$\begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix}$$

Recalling that:

$$u_h^1(\sigma^2) = e_h^T A \sigma^2, \qquad \text{and} \qquad u_h^2(\sigma^1) = e_k^T A \sigma^1$$

We find:

$$\begin{aligned}
u_h^1(\sigma^2) = e_h^T A \sigma^2 = 3\sigma_h^2, \quad u_k^1(\sigma^2) = e_k^T A \sigma^2 = 4\sigma_h^2 + 1 \\
u_h^2(\sigma^1) = e_h^T A \sigma^1 = 3\sigma_h^1, \quad u_k^2(\sigma^1) = e_k^T A \sigma^1 = 4\sigma_h^1 + 1
\end{aligned}$$

Resulting in the characteristic replicator dynamic for this payoff matrix *(player 1 playing strategy h)*

$$\begin{aligned}
\dot{\sigma_h}^1 &= \sigma_h^1 \sigma_k^1 (3\sigma_h^2 - (4\sigma_h^2 + 1)) \\
&= \sigma_h^1 \sigma_k^1 (-\sigma_h^2 - 1) \\
&= -\sigma_h^1 \sigma_k^1 (\sigma_h^2 + 1)
\end{aligned}$$

Setting this equal to 0 to find the fixed points we see:

$$\begin{aligned}
0 &= -\sigma_h^1 \sigma_k^1 (\sigma_h^2 + 1) \\
\sigma_h^{1*} &: \sigma_h^{1*} = 0 \text{ or } \sigma_k^{1*} = 0 \\
\sigma_h^{1*} &: \sigma_h^2 = -1
\end{aligned}$$

It is not physically possible for a probability to be negative so we reject the case where $\sigma_h^2 = -1$. Leaving us with fixed points where $\sigma_h^1 = 0$ or $\sigma_k^1 = 0$ Putting solely in terms of strategy h, $\sigma_h^{1*} = 0$ or $(1 - \sigma_h^1) = 0 \rightarrow \sigma_h^{1*} = 1$.
Graphing computationally in python we see that $\sigma_h^{1*} = 1$ is unstable and $\sigma_h^{1*} = 0$ is stable. The proportion of strategy h will eventually be 0. This agrees with our qualitative discussion at the beginning of this paper when we showed that the Nash Equilibrium of this matrix was to defect. This result can be repeated for each player playing every strategy to get a complete continuous ODE system that models the trajectory of strategies played. This results in a vector field of movement in the game in figure 1.

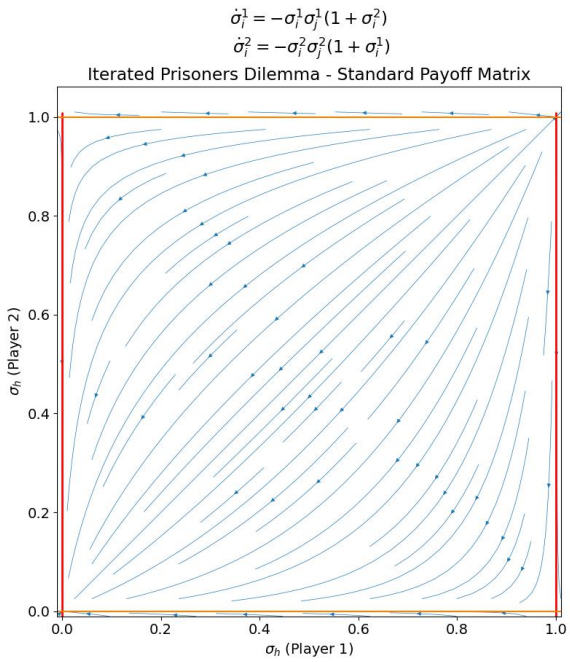


Figure 1: Vector Field of Standard Prisoners Dilemma

2.6 Weibull Example Replicator

In Weibull's *Evolutionary Game Theory* he discusses replicator dynamics towards the end of his book. In there he provides an example of another finite positive sum symmetric system that behaves with a Nash equilibrium. This dynamic is given by:

$$\dot{\sigma}_h^i = -\sigma_h^i(1-\sigma_h^i) \rightarrow \dot{\sigma_h^j} = -\sigma_h^j(1-\sigma_h^j)$$

An example matrix of which could be:

$$\begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$$

As in the standard dynamic, this system displays the same phase portrait. We have fixed points at 0 & 1. Computationally we see 1 as unstable and 0 as stable. These equivalent vector fields generating from 2

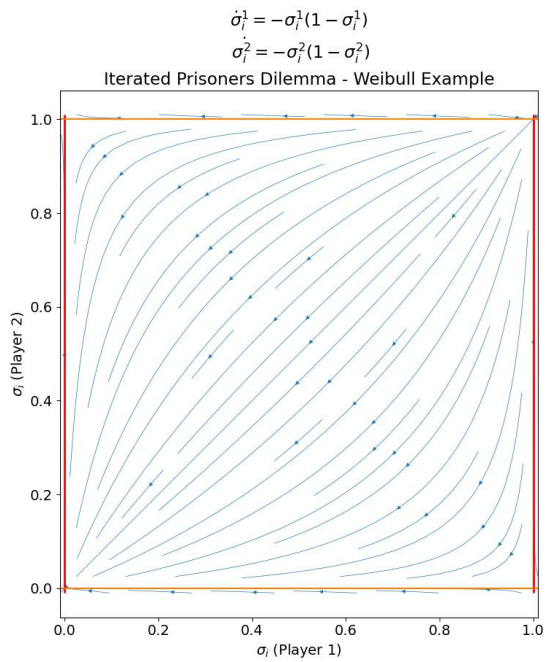
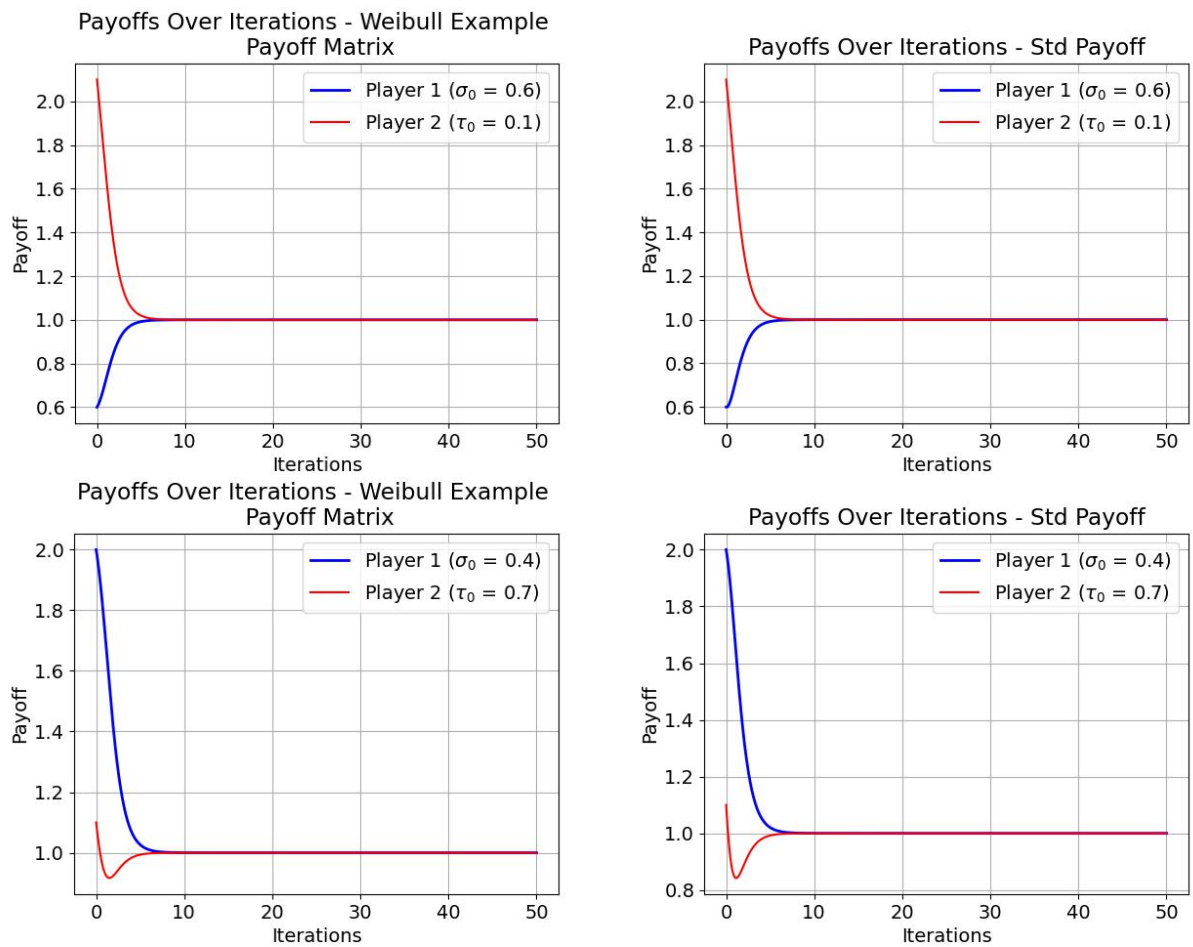


Figure 2: Vector Field of Standard Prisoners Dilemma

separate replicator dynamics generated from 2 different payoff matrices corroborates the Nash Equilibrium theory.

2.6.1 Weibull Example Payoffs

The expected payoffs as generated by this dynamical system support the calculation of expected payoff. A few examples follow.



In both replicator examples the players trend to each getting a payoff of 1. In our case this is the "defect" option, our Nash Equilibrium.

Standard Payoff Matrix Replicator Function in Python

```
def ipd(t, z):  #from [3,0],[5,1] Standard Payoff
    p1_sigma, p2_sigma = z
    p1_sigma_dot = -p1_sigma*(1-p1_sigma)*(1+p2_sigma)
    p2_sigma_dot = -p2_sigma*(1-p2_sigma)*(1+p1_sigma)
    return [p1_sigma_dot, p2_sigma_dot]
```

Weibull Example Payoff Matrix Replicator Function in Python

```
def ipd(t, z): #from [2,0],[3,1] Weibull Payoff Matrix
    p1_sigma, p2_sigma = z
    p1_sigma_dot = -p1_sigma*(1-p1_sigma)
    p2_sigma_dot = -p2_sigma*(1-p2_sigma)
    return [p1_sigma_dot, p2_sigma_dot]
```

3 Summary & Next Steps

To say the study of Game Theory is a broad subject matter would be an understatement. This research paper constitutes a very introductory and basic examination of the fundamentals. We reviewed single-shot vs. iterated prisoners dilemma. This was followed by a discussion of the replicator dynamic and its use in deriving continuous systems of ordinary differential equations for use in plotting game dynamics. Next areas of study could include anything from increasing the dimension of strategies available to increasing player count or finding fixed points and calculating their stability. This study of game theory was enlightening in advising complexity and depth of the use of game theory in the modeling of numerous fields of research.

Cited Work

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[1]

[1] Please note that Claude Opus 4.5 was used in the formatting and grammar of this paper and accompanying presentation.

Appendix A

Other examples of Notation for "payoff" functions.

Symbol	**Typical Meaning**	**Context**
$u(x, y)$	Game Payoff	Game Payoff, payoff matrices
$W(x, y)$	Evolutionary Fitness	Population Dynamics, evolution
$F(i\|\|x)$	Fitness of strategy$i$	Replicator Dynamics
$\pi(x, y)$	Payoff/Profit	Economics, game theory